

Leonardo Bove

Quantum Engineer, SQMS Fermilab
Ph.D. Student in Computer Science

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EDUCATION

Ph.D. in Computer Science

Northwestern University

Evanston, IL, USA

Sep. 2026 – Present

- Advisors: Professor Nikos Hardavellas, Professor Kaitlin Smith
- Research interests: compilers, algorithms, and infrastructure for fault-tolerant quantum computing (FTQC), quantum error correction, and quantum control, with a focus on the 3D SRF cavity bosonic mode qudit approach

M.Sc. in Computer Science

Northwestern University

Evanston, IL, USA

Sep. 2026 – Present

M.Sc. in Electronic Engineering

University of Pisa

Pisa, Italy

Sep. 2023 – Oct. 2025

- Specialization: Embedded Systems & Mechatronics
- Thesis title: "*Superconducting qubit readout and control system based on FPGA and development of a pulse sequencer*"
- Thesis advisors: Professor Massimo Macucci, Professor Stefano Di Pascoli, Dr. David Van Zanten
- Degree grade: 110/110 *summa cum laude* (highest distinction)

B.Sc. in Electronic Engineering

University of Pisa

Pisa, Italy

Sep. 2020 – Jul. 2023

- Thesis title: "*Dispersive readout of the Transmon qubit*"
- Thesis advisor: Professor Massimo Macucci
- Degree grade: 110/110 *summa cum laude* (highest distinction)

RESEARCH EXPERIENCE

Quantum Engineer

SQMS, Fermilab

Batavia, IL, USA

Jan. 2026 – Sep. 2026

- Calibration of SRF cavities using Quantum Machines control systems (*OPX100* and *OPX+* with *Octave*).
- Developed automatic calibration routines using *Qualibrate*.
- Calibrated the *Novera* Rigetti QPU using the *OPX1000* platform.
- Developed the infrastructure for auto-calibrating the *Novera* QPU and making it accessible on the cloud for users.

Graduate Research Fellow in Quantum & Electronic Engineering

Department of Information Engineering, University of Pisa

Pisa, Italy

Sep. 2025 – Present

Recipient of a 3-month research fellowship titled "*Measurement of relaxation and decoherence times in superconducting qubits*".

- Characterization of multiple qubits using frequency multiplexing strategies. Performed the first qubit measurements ever conducted within the engineering department.

- Management of the laboratory's dilution refrigerator infrastructure.
- Microwave path engineering: component sizing and selection, and system characterization using RF instrumentation (VNA, spectrum analyzer). Maximizing the available SNR in the readout chain without relying on quantum-limited amplifiers.
- Developed FPGA-based qubit control and readout using the *QICK* platform, adapting it to our experimental setup. Achieved comparable performance to off-the-shelf commercial qubit controllers while significantly reducing hardware costs.
- Established a remote-access infrastructure for qubit experiments (*JupyterLab server*).
- Trained and supervised master's students in laboratory research, with a focus on qubit calibration procedures.

Master Thesis Project

SQMS, Fermilab

Batavia, IL, USA

May 2025 – Jul. 2025

- Designed and analyzed *QPCB*, a custom pulse frequency conversion board, using microwave EM simulations (*ADS Keysight*).
- Engineered *Qubase*, a high-level Python pulse sequencer for the *QICK* FPGA platform, leveraging *rustworkx* for sequence graph analysis.
- Applied Qubase to characterize 2D and 3D superconducting qubits and support advanced quantum experiments (e.g. parallel fast-decay detection sequence).
- Automated calibration workflows for high-throughput measurements.

PROFESSIONAL EXPERIENCE

Chief Technology Officer

E-Team Squadra Corse, FSAE team

Pisa, Italy

Sep. 2023 – Sep. 2024

- Directed the development of the University of Pisa's EV race car (*Electronics & Software Divisions*) for the 2023–2024 season.
- Achieved a significant performance improvement at Formula Student Germany 2024, finishing 11 positions higher in the overall ranking compared to the previous year, and ranking as the only Italian team to successfully complete the endurance event.
- Approved and validated engineering projects.
- Oversaw high-voltage battery systems.
- Administered and maintained the team's GitLab server. Coordinated CI pipelines. Reviewed pull requests and issue workflows.
- Fostered team cohesion through structured team-building initiatives.
- Organized personnel logistics and supervised technical activities.

Embedded Software Developer

Sintonica s.r.l.

Navacchio (PI), Italy

May 2023 – Sep. 2023

- Developed firmware drivers for TFT LCD displays on a custom embedded OS using Infineon PSoC ARM microcontrollers.
- Contributed to the layout and enhancement of the company's development kit PCB, integrating Cypress and nRF PSoC platforms.

PCB Designer and Embedded System Developer

E-Team Squadra Corse, FSAE team

Pisa, Italy

Sep. 2022 – Sep. 2023

- Led development of embedded software for onboard PCB systems.
- Implemented components of the Vehicle Control Unit using the FreeRTOS real-time OS.
- Engineered a multi-architecture bootloader (ARM/AVR) with CAN bus support, enabling rapid firmware updates for inaccessible microcontrollers in time-critical scenarios.

- Conducted unit and integration testing of firmware modules. Co-developed a virtualization framework enabling *Model-in-the-loop* and *Software-in-the-loop* testing of firmware prior to deployment on physical hardware.

SKILLS

Software Development Tools & OS

Git, Linux, Windows

Programming Languages

C/C++, Python, Verilog, VHDL, MATLAB, Bash, Assembly

Quantum Computing

Superconducting qubit & bosonic cavity mode qudit characterization, Quantum Machines (OPX, OPX+, OPX1000, Octave), Qualibrate, Qiskit, QuTiP

FPGA Design

Vivado, Quartus, ModelSim

Embedded Systems

STM32CubeIDE, PSoC Creator, Simulink Model-Based Design

Analog Design

SPICE, ADS, KiCad, Altium

MEMS Design & Simulation

COMSOL Multiphysics

Microcontroller Architectures

AVR, ARM

PCB Design

KiCad, Altium

Lab & Fabrication

VNA, soldering, prototyping, Autodesk Fusion 360, 3D printing

LANGUAGES

Italian Native speaker

English C2 level (Cambridge *Certificate in Advanced English*, 2020)

German B2 level (Linguistic Center *CLI* University of Pisa, 2022)

Spanish B1 level

French A2 level

INVITED TALKS & PRESENTATIONS

"Superconducting qubit readout and control system based on FPGA and development of a pulse sequencer", Quantum Device Lab, Department of Physics, ETH Zürich, Zürich, Switzerland, July 23, 2025.

GRAD COURSE PROJECTS

SpaceFibre PLL

Pisa, Italy

University of Pisa, *Wireless Integrated Circuits*, Professor Daniele Rossi

Sep. 2022 – Sep. 2023

- Modeled and simulated a SpaceFibre-compatible 6.25GHz PLL, using the SG25H4 0.25 μm SiGe BiCMOS technology.
- Evaluated system stability and performance metrics.

Handwritten Digit Recognition

Pisa, Italy

University of Pisa, *Digital System Design*, Professor Roberto Saletti

Jan. 2025 – Feb. 2025

- Engineered a neural-network-based handwritten digit recognizer on the Altera DE10-Lite (MAX10 FPGA).
- Built a Python digital twin for training and quantization.

- Implemented custom Verilog drivers for touchscreen control and LT24 visualization.

Dual Axis Accelerometer

Pisa, Italy

University of Pisa, *Sensor and Microsystem Design*, Professor Massimo Piotto Dec. 2024 – Jan. 2025

- Simulated a dual-axis MEMS accelerometer with T-shaped beams using COMSOL.
- Analyzed device performance and application range, benchmarking against lumped-element models.

Rubik's Cube Automatic Solver

Pisa, Italy

University of Pisa, *Mechatronic Systems Design*, Professor Roberto Di Rienzo Nov. 2024 – Dec. 2024

- Designed a servo-actuated Rubik's Cube solver robot controlled by an S32K144EVB.
- Developed the system using Simulink Model-Based Design.
- Built a digital twin of the complete robotic system.

HONORS & AWARDS

Industry 5.0 Excellence Learning Path

2025

FoReLab project, University of Pisa

Specialized training program designed to develop advanced skills in next-generation industrial technologies by integrating research, innovation, and hands-on projects aligned with the principles of Industry 5.0.

Industry 4.0 Learning Path

2025

University of Pisa

Training program that provides practical and theoretical skills in digital manufacturing, smart automation, and modern industrial technologies aligned with the principles of Industry 4.0.

EXTRACURRICULAR ACTIVITIES

Private Tutoring

Pisa, Italy

2023 – 2025

- Conducted weekly one-to-one tutoring sessions for more than 5 high school students.
- Provided support across multiple subjects, including Computer Science (e.g., graph theory, computer networks, Java), Electronics (digital and analog circuits), Physics, and Mathematics.
- Achieved an average 30% improvement in students' grades.

Violin

Montepulciano, Italy

"Hans Werner Henze" Institute of Music

2009 – 2021

- Individual lessons and orchestral performances with the *Poliziana Orchestra* at the annual *Cantiere Internazionale d'Arte di Montepulciano*, in addition to periodic symphonic music concerts.
- Recipient of three merit-based scholarships awarded by the *Cantiere Internazionale d'Arte di Montepulciano* foundation.